Where the Bias Gets In: Navigating Human Bias as an AI Leader

Wenyu (Daniel) Du · Indiana Wesleyan University · August 2026

As a software engineer who has taken an LLM system from prototype to production, I have learned that bias rarely enters an AI system through the algorithm alone. It enters through the chain of human judgments surrounding the algorithm — five doors, in practice: which data is selected, how it is labeled, how the algorithm is designed, how the system is tested and evaluated, and how its results are interpreted and used. Because those judgments are human, leading AI/ML integration is ultimately an exercise in change leadership as much as engineering.

Personal Value Statement

My core value as a technical leader is deliberate humility: the working assumption that my own judgment is itself a potential source of bias, and that fairness must be engineered through explicit process, diverse input, and honest measurement rather than assumed from good intentions. None of the strategies below assumes I am unbiased. All of them assume the opposite — and build the process that catches what I will inevitably miss.

Strategy 1: Treat Data Selection and Labeling as Ethical Decisions

In a production text-to-SQL system I led, our internal database contained roughly 1,500 tables, which made training on the full schema infeasible. We narrowed the training scope to the tables and relationships we judged most relevant, and we fed verified correct SQL back into the model as positive training examples. Both decisions were technically sound, but each embedded human judgment: "relevant" meant relevant to the workflows my team saw most often, and "verified" meant verified by us. If our curation over-represented the query patterns of the departments we worked with most closely, the system would serve those users well while quietly underperforming for everyone else. My strategy is to make these choices visible — documenting selection criteria explicitly and inviting stakeholders outside the core engineering team to review what was included and, more importantly, what was left out. The practice Gebru et al. (2021) call a *datasheet for a dataset* formalizes exactly this: a short, structured record of where the data came from, who is represented in it, and what it should not be used for.

Strategy 2: Guard Evaluation Against Confirmation Bias

Confirmation bias — our tendency to notice what supports what we already believe (Kahneman, 2011) — has a precise engineering form: engineers naturally test with the queries they expect to work, and a single aggregate accuracy number can conceal exactly who a system is failing. Buolamwini and Gebru (2018) demonstrated how consequential this is, finding that commercial vision systems with strong overall accuracy performed far worse for darker-skinned women — a gap that stayed invisible until someone disaggregated the results. I address this by building evaluation sets from real user queries across different departments and roles, and by reporting performance disaggregated by user group rather than accepting a comfortable average.

Strategy 3: Design Feedback Loops with Bias in Mind

In our system, user-confirmed SQL became future training data, which steadily improved quality — but a loop like this can also amplify bias, because the users who give the most feedback pull the model further toward their own patterns. My strategy is to actively solicit input from quieter and less technical user groups and to balance the feedback data before it is used for training, so the model does not evolve to serve only its most vocal users.

Strategy 4: Lead the Human Side of Change

Navigating bias is central to leading AI adoption. When we rolled out the text-to-SQL tool, I observed two opposite biases among users: some accepted generated SQL uncritically because "the AI wrote it," while others dismissed the entire system after a single wrong answer. Neither reaction reflected the tool's actual reliability. As a change leader, my strategy is to calibrate trust deliberately — communicating the system's limitations transparently, training users to verify outputs rather than assume them, and treating every reported error as a contribution instead of a complaint. Modeling that posture myself, and building the psychological safety for any team member to challenge a dataset choice or an evaluation result, is in my experience the cheapest and most effective bias-mitigation tool a technical organization has.

References

Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. *Proceedings of Machine Learning Research*, *81*, 77–91.

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, *64*(12), 86–92.

Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.

Disclosure: Claude (Anthropic) assisted with drafting and formatting. The value statement, production examples, and positions are my own.